Poster project: one question, one poster

CE/EL/MC/CV/GL/GM/ES 361 Probability and Statistics • First Semester, 2026/2027

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What this is

One project for the whole course, presented as a **poster**. Your group takes one question from your own field of engineering, collects or finds the data, and answers it using the statistics of this course: a description, a probability model, an interval, a test, and a relationship.
Groups of five or six, from the same programme where possible. **Poster session: the week of Monday 7 to Friday 11 December 2026.** Your group's day, time and venue within that week will be announced in class and posted on the course page.
There is no written report. The poster and your answers at it are the whole assessment, and both are marked. Worth 100 marks.

What the poster must contain

Eight things, in this order. The order is the argument, and a poster that jumps straight to a conclusion loses most of the marks.

Section	Chapter	What we must see
The question	1	one question, narrow enough to answer, stated in one sentence
The data	1	how you sampled, how many, and who or what is missing from it
Description	1	centre, spread and shape of your key variable, with one figure
A model	2 to 5	a distribution chosen from the mechanism, with its parameters estimated from your data
An interval	6	a confidence interval for a mean or a proportion, in the units of the problem
A test	7	one hypothesis, stated before you looked, tested and reported with its p-value
A relationship	8	a scatter plot with a fitted line, the slope in real units, and r or R^2
Limits	all	what your data cannot support, in specific terms

The poster itself

- **A1 size, portrait.** Printed, or drawn by hand on A1 paper. A hand drawn poster is marked exactly like a printed one: neatness counts, money does not.
- **Readable from one and a half metres.** Titles about 70 pt, section headings about 36 pt, body text no smaller than 24 pt. If you can read it comfortably on your laptop at 25% zoom, it is big enough.
- **At most 600 words.** The figures and the numbers do the work. Long paragraphs are left unread by every visitor to the session.
- **Two or three figures, large.** Axes labelled with units, sample size stated, no three dimensional charts, no pie chart with more than three slices, and nothing that depends on colour alone.
- **Every number on the poster must be reproducible** from the data and code you submit. A number nobody can reproduce scores zero, however correct it looks.

Choosing a question

Pick something you can actually measure in the time available. Good projects from previous years were small: the compressive strength of blocks from two suppliers in one yard; waiting times at one campus printer; moisture content of samples from one stockpile; the time between truck arrivals at one gate; the resistance of 40 resistors from one batch.

Thirty honest observations you collected beat three thousand you downloaded and cannot explain. If you use existing data, you must be able to say who collected it, when, how, and who is missing from it. If you cannot answer those four questions, choose different data.

Two groups in the same class may ask the same question, but not with the same data.

What to submit, and when

1. **The poster**, as a PDF, or as a clear photograph if it is hand drawn, uploaded before your session begins.
2. **Your data and your working**, as one file: the dataset as you used it, plus the spreadsheet or script that produced every number on the poster.
3. **A one page memo**, which lists the group members and who did what, where the data came from, and the hypothesis you fixed before analysing.

Nothing is accepted after your session has started, and a group that has not uploaded does not present.

How it is marked

Component	Marks
The question and the data: narrow, answerable, properly sampled, honest about what is missing	20
The statistics: interval, test and relationship correctly computed and correctly read	30
Figures: chosen for the comparison, labelled, legible, honest	15
Limits: a specific account of what the data cannot support	10
Poster craft: layout, legibility at one and a half metres, within the word limit	10
At the poster: your two minute walkthrough and your answers to questions	15

The first five rows are a group mark. The last row is individual: I will ask each member of the group a question, and "my group mate did that part" scores zero for that question. Divide the work however you like, but everybody must understand the whole poster.

At the poster session

I will stop at each poster twice. The first time, one member gives a two minute walkthrough. The second time, I pick a member and ask two or three questions of this kind:

- How did you choose these thirty samples, and what would a different choice have done to your interval?
- Why that distribution and not another one?
- Your interval is this wide. What would make it narrower, and what would that cost?
- What exactly does your p-value mean? Say it without using the word "chance".
- Your line predicts this value at $x = 50$. Would you sign for that number?
- Which number on this poster are you least confident about?

Using AI tools

Allowed, and must be declared in one line at the bottom of the poster: which tool, and what for. That line costs no marks. An undeclared tool is academic dishonesty and goes to the Department. Be warned that these tools produce confident statistical sentences that are subtly wrong, and the questions above are designed to find exactly that. If you cannot explain a sentence on your poster in your own words, take it off the poster.

Timetable

Step	What happens
Register	group members and your question, in one paragraph, handed to me in class
Check-in	bring your data and your planned hypothesis, before you analyse anything
Upload	poster, data, working and memo, before your session
Poster session	the week of 7 to 11 December 2026, slot announced in class

Advice, from the marker

1. **Shrink the question until it is answerable in four weeks.** This is where most marks are won and lost.
2. **Collect the data early.** Every group that leaves it late ends up analysing whatever was easy to get, and it shows.
3. **Write the limits section first.** Drafted in week one, it tells you what to fix while there is still time to fix it.
4. **Make one figure excellent** rather than four that are adequate.
5. **Rehearse the two minutes standing up**, in front of the poster, with somebody who is not in your group.